Measurement of Proton-Tagged Charged-Current Single-Charged-Pion Production on Argon in the NuMI Beam with the MicroBooNE Detector

Companion to the inclusive analysis note

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Abstract

The inclusive CC$\pi^{\pm}$ measurement of the companion note is repeated on the subsample in which a proton candidate with $p_p > 0.3\,\mathrm{GeV}/c$ is also reconstructed. All results are extracted from pseudo-data. The proton gives access to the visible hadronic system: the proton momentum and direction, the pion–proton opening angle, a calorimetric hadronic-mass proxy and the transverse kinematic imbalance between the muon and the πp system. The cross sections are averaged over the $\nu_\mu + \bar{\nu}_\mu$ flux integrated above 60 MeV and extracted in the FHC, RHC and combined configurations with the Wiener-SVD machinery of the inclusive analysis, each with its additional-smearing matrix and covariance. The one-bin total cross section, $d\sigma/d\theta_p$ and $d\sigma/d\theta_{\pi p}$, and the combined-configuration $d^2\sigma/d\theta_p\, d\delta p_T$ are also obtained. The one-bin total carries a 41–43% uncertainty; ensembles of pseudo-experiments give pull widths of 0.92–1.09.

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1 Introduction

The inclusive analysis [11] measures charged-current single-charged-pion production with any number of nucleons in the final state. Requiring in addition a reconstructed proton gives access to the visible hadronic system and to observables that the inclusive measurement cannot provide.

These results address three questions. The one-bin total measures the CC$\pi^{\pm}$ cross section in the final states with at least one proton above 0.3 GeV/c, which depends on the resonance decay, on nucleon final-state interactions and on the nuclear model: among the generators the ratio of the proton-tagged to the inclusive FHC total ranges from 0.52 (NuWro) to 0.65 (GiBUU), with 0.61 for the tune. The proton-tagged totals themselves span 0.50 (GENIE) to 0.78×10^{-38} cm^2/Ar (NEUT) in FHC, against a 42–47% uncertainty on the measured total, so the two extreme predictions differ by about one standard deviation of the measured total. The proton polar angle θ_p measures the proton direction, and the pion–proton opening angle $\theta_{\pi p}$ how the momentum of the hadronic system is shared between the pion and the proton, which are complementary to the transverse-kinematic-imbalance (TKI) observables of the summed system. The two-dimensional $\theta_p \times \delta p_T$ relates the proton direction to the transverse imbalance.

The proton requirement changes more than the sample size. The signal definition acquires a proton threshold and a leading-proton convention. The reconstruction acquires a proton candidate that is the leading true proton in 82% of selected signal events. Two observables, p_n and the hadronic-mass proxy, rely on a single-nucleon residual-mass prescription written for quasi-elastic kinematics and not yet validated for pion production. The detector uncertainty, evaluated on the smaller sample, reaches 75% in the sparsest extractions. The inclusive measurement carries none of these, and the two analyses are therefore documented separately.

2 Signal definition and selection

Signal. The signal is the inclusive signal of Sec. 6 of the inclusive analysis note with at least one true proton above 0.3 GeV/c; the muon and pion phase space is unchanged. The truth-level proton observables use the leading proton, the one of highest momentum.

Selection. The ten inclusive cuts of Sec. 8.2 of the inclusive analysis note are applied unchanged, followed by an eleventh cut that identifies the proton:

Requirement	Threshold	Description
Candidate set	–	track-like primaries other than the muon and pion candidates
Proton likeness	LLR PID < 0.05	the most proton-like of the set is the candidate
Containment	end point in FV	range-based momentum is meaningful only if contained
Momentum	$p_p \geq 0.3$ GeV/c	tracking threshold; below it protons are not reconstructed reliably

The proton momentum is computed from the proton-hypothesis kinetic energy of the candidate track, which is obtained from its range, as $p_p = \sqrt{T_p^2 + 2 m_p T_p}$. On truth-matched FHC simulation it underestimates the true momentum by 9% on average, with a resolution of 21%; both are part of the response matrix.

Performance. At full exposure the selection has an efficiency of 12.1% and a purity of 52% in FHC (50% in RHC, 50% combined), against 18.3% and 57% for the inclusive selection. Relative to the inclusive final cut evaluated under the proton-tagged signal definition, the proton requirement rejects 68% of the neutrino background for a 31% loss of signal, raises the purity from 33% to 52% and improves $S/\sqrt{B}$ from 15.9 to 19.2 (FHC). It also rejects cosmic rays: a contained proton above 0.3 GeV/c reduces the beam-off background by a factor 3.2–3.4 at the final cut and by up to

six in the CC0π control region (Sec. 20.7 of the technical supplement). The cut-flow by stage and category is Table 11 of the technical supplement (stacks in Fig. 6 of the technical supplement); it uses the per-run exposure scaling of the inclusive cut-flow (Table 11 of the inclusive analysis note), and its final row gives the counts of Table 5.

2.1 What the proton candidate is

On selected signal events the proton candidate is the leading true proton in 81.7% of cases, some true proton in 87.4%, and not a proton in 12.6% (10 105 simulated events). The wrong candidates are mainly charged pions or other tracks taken for the proton.

The effect of a wrong candidate was measured for the two angular observables. A correct candidate reconstructs the angle precisely, with an interquartile range of the residual of 0.073 rad in θ_p and 0.124 rad in $\theta_{\pi p}$. A wrong candidate gives an angle unrelated to the true one: the interquartile range is 1.0–1.4 rad, and three quarters of those events have residuals beyond 0.3 rad. The core resolution averaged over all events is only a quarter to a third wider than that of correct candidates, but the migrations between bins come from the tails: residuals beyond 0.3 rad occur in 18.9% of θ_p events, and 74% of them are wrong candidates (25.9% and 51% for $\theta_{\pi p}$). The candidate purity is therefore the main source of bin-to-bin migration in θ_p, whose smallest migration diagonal is 0.69 in FHC and 0.68 in RHC at the adopted edges.

The same wrong-candidate rate applies to $W_{\pi p}$, W_{had} and the TKI observables, which use the same candidate; the attribution of the migrations was measured for the angles only. The multi-π control region of the inclusive analysis receives otherwise-signal events whose proton fails the proton-likeness requirement, since the pion count uses the same PID cut. It constrains the proton→pion direction, not the pion→proton direction that produces the 12.6%, and it is the region with the largest signal content. A control region that inverts the proton PID at exactly one pion candidate would separate the two directions (Sec. 20.6 of the technical supplement).

3 Observables and binning

Definitions. All observables are built from the three candidate tracks with reconstructed quantities only: the range or multiple-scattering momentum of the muon, the range momentum of the pion, the range kinetic energy of the proton and the track directions. Angles and transverse quantities are taken about the neutrino direction of Sec. 4.2 of the inclusive analysis note, the same convention as the inclusive analysis and the generator predictions.

- The proton momentum p_p is that of the candidate; at truth level it is the momentum of the leading proton.
- The πp invariant mass, $W_{\pi p} = \left[(E_\pi + E_p)^2 - (\vec{p}_\pi + \vec{p}_p)^2\right]^{1/2}$, is measured directly and needs no neutrino direction.
- The calorimetric hadronic-mass proxy, $W_{\mathrm{had}} = \left[m_p^2 + 2m_p(E_{\mathrm{cal}} - E_\mu) - Q^2\right]^{1/2}$, is built from the reconstructed calorimetric energy E_{cal} and Q^2.
- The TKI observables are the transverse-momentum imbalance δp_T, its angles $\delta\alpha_T$ and $\delta\phi_T$ [8], and the inferred nucleon-momentum proxy p_n [9], computed from the muon against the summed $(\pi + p)$ system with the STV tool of the framework after rotating both into the frame in which the neutrino defines the z axis.
- The proton polar angle and the pion–proton opening angle,

  $$\theta_p = \arccos\left(\hat{p}_p \cdot \hat{\nu}\right), \qquad \theta_{\pi p} = \arccos\left(\hat{p}_\pi \cdot \hat{p}_p\right),$$

  of which $\theta_{\pi p}$ is an angle between two reconstructed tracks and does not depend on the neutrino direction.

The five muon and pion observables of the inclusive analysis (p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) are measured again on the proton-tagged phase space. The formulae and the conventions of the TKI tool are given in Sec. 9.1 of the technical supplement.

The TKI observables and W_{had} are built from the summed hadronic system and are insensitive to how its momentum is shared between the pion and the proton; $\theta_{\pi p}$ measures that sharing. θ_p is the proton direction itself, of which the other observables use only derived quantities.

Binning. Every binning meets the 0.50 migration criterion of the inclusive analysis with at least 50 selected FHC signal events per bin (Sec. 7 of the technical supplement; Table 1): p_p in three bins split at 0.42 and 0.60 GeV/c with an open upper bin, W_{had} in two regions split at 1.18 GeV/c^2, $\delta\phi_T$ in three bins, δp_T, $\delta\alpha_T$ and p_n in two, and θ_p and $\theta_{\pi p}$ in four each, at edges that maximise the smallest migration diagonal. The muon and pion observables use the inclusive edges, except p_π, which has two regions because the three inclusive regions fail the criterion in this sample, and $\cos\theta_\pi$, which has four bins. The RHC and combined samples use the FHC binnings.

Table 1: Bin edges of the proton-tagged observables, from the result files. The binnings meet the 0.50 migration criterion (Sec. 7 of the technical supplement) and are the same in the three configurations.

Observable	Bins	Edges
p_μ (GeV/c)	6	0.15, 0.3, 0.425, 0.575, 0.825, 1.25, 3*
p_π (GeV/c)	2	0.175, 0.205, 1*
$\cos\theta_\mu$	8	-1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1
$\cos\theta_\pi$	4	-1, -0.1, 0.35, 0.75, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6
$W_{\pi p}$ (GeV/c^2)	6	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9*
W_{had} (GeV/c^2)	2	0, 1.18, 2.74*
δp_T (GeV/c)	2	0, 0.3, 2.5*
$\delta\phi_T$ (deg)	3	0, 22.5, 75, 180
$\delta\alpha_T$ (deg)	2	0, 120, 180
p_n (GeV/c)	2	0, 0.375, 2*
p_p (GeV/c)	3	0.3, 0.42, 0.6, 1.5*
θ_p (rad)†	4	0, 0.44, 0.785, 1.162, π
$\theta_{\pi p}$ (rad)†	4	0, 0.911, 1.414, 1.916, π

† Released as a Secondary result (§6.2).

* Numerical plotting boundary only: the upper truth bin is open-ended, and a density drawn over the conventional width represents the bin integral.

4 Extraction and uncertainties

Each observable is extracted as in the inclusive analysis: the selected spectrum is scaled per run to the recorded exposure, the neutrino, beam-off and dirt backgrounds are subtracted, and the result is unfolded with the Wiener-SVD method [1] and the response matrix of the proton-tagged signal (Sec. 11 of the inclusive analysis note). Every result is provided with its additional-smearing matrix A_C, through which a prediction must be passed before comparison, and with its covariance by source (Sec. 14.4 of the inclusive analysis note). The combined configuration is an independent extraction with its own response and A_C, not an average of FHC and RHC.

The covariance contains the flux, cross-section, re-interaction, detector, POT, target-count and statistical terms of the inclusive analysis. The detector term ranges from 6.8% to 47.5% (median 19.2%) against 12.1–24.9% for the inclusive results. It is of the same size in most extractions, as expected for the same detector, and largest in the sparsest ones (RHC p_n, p_π

and W_{had}, 41–48%; the six-bin $W_{\pi p}$, not reported, is excluded), where the unfolding amplifies the variations. The flux term of the one-bin total is 35.1% in FHC. As in the inclusive analysis (Sec. 12.2 of the inclusive analysis note), most of it is the 17.7% flux normalisation multiplied by $1 + B/S$ through the subtraction of the flux-dependent background, the 319 simulated neutrino-background events against 349 signal events of Table 5 (its other 9 background events are beam-off and dirt): $17.7\% \times (1 + 319/349) = 33.9\%$. The propagated flux universes also change the spectrum shape, which gives the rest.

4.1 Muon momentum scale

The muon momentum scale is varied as in the inclusive analysis (Sec. 12.3 of the inclusive analysis note): in the pseudo-data alone, every muon whose momentum is measured by multiple Coulomb scattering is scaled by 1 ± 0.05, while the simulation, the response and the selection stay nominal, and each variation is unfolded with the nominal response. The half-difference of the two results is taken as a fully correlated term. In this sample the muon momentum also enters W_{had}, δp_T, $\delta\alpha_T$ and p_n, which are therefore recomputed from the scaled muon for every event with a proton candidate; with a scale of one the recomputed values reproduce the stored ones exactly. $\delta\phi_T$ depends only on the muon direction, and $W_{\pi p}$, p_p, p_π and the angles do not use the muon momentum, so their term is zero.

Table 2 gives the result. The largest term is in the open highest p_μ bin, 26% of the bin value and 0.69σ of its uncertainty in FHC; the lower p_μ bins move by 8–16%, as in the inclusive analysis. For the derived observables the term is at most 0.45σ: it is largest in the lower W_{had} region (18–50%, 0.26–0.45σ), in the upper p_n bin (14–24%) and in the lower $\delta\alpha_T$ bin (15–18%), and it is below 0.2σ for δp_T. Adding the term raises the per-bin total uncertainty by at most 21% of itself for p_μ (FHC) and by at most 10% for the derived observables. The term is part of the official covariance of every result. Every uncertainty quoted in this note includes it; the closure χ^2 and the robustness biases are quoted with the framework covariance.

Table 2: Muon momentum-scale term (±5% on the MCS-measured muons of the fake data) for the proton-tagged observables that use the muon momentum: the change of the integral for the upward and downward scale, and the largest per-bin half-difference as a fraction of the bin value and in units of the total uncertainty of that bin.

Config	Observable	up / nominal	down / nominal	largest half-diff.	in σ
FHC	p_μ	0.967	1.031	26.4%	0.69
	W_{had}	1.019	0.984	17.6%	0.26
	δp_T	0.973	1.004	3.8%	0.12
	$\delta\alpha_T$	0.932	1.077	16.3%	0.37
	p_n	0.994	1.004	23.5%	0.28
RHC	p_μ	0.993	1.021	14.5%	0.35
	W_{had}	1.059	0.972	50.1%	0.45
	δp_T	0.946	1.014	4.1%	0.17
	$\delta\alpha_T$	0.923	1.089	15.1%	0.42
	p_n	0.999	1.001	22.2%	0.20
Combined	p_μ	0.994	1.037	19.4%	0.55
	W_{had}	1.038	0.978	40.8%	0.45
	δp_T	0.938	1.013	3.8%	0.18
	$\delta\alpha_T$	0.925	1.086	17.7%	0.45
	p_n	0.976	1.013	13.8%	0.30

5 Validation

The validation follows that of the inclusive analysis (Sec. 13 of the inclusive analysis note). Every proton-tagged extraction closes under the nominal model, with $\chi^2 \leq 7.28/4$ (RHC $\cos\theta_\pi$) and integral ratios of 0.91–1.21; the two-dimensional result has cell ratios of 0.62–1.32 (§5.1). Ten extractions, the differential results among them, were repeated on 100 pseudo-data sets each, with pull widths of 0.92–1.09 (§5.2). Pseudo-data from two reweighted interaction models were unfolded for the proton-tagged observables in FHC and for the two-dimensional result in the combined configuration (§5.3): the low-imbalance δp_T and p_n bins move by -1.08σ and -1.04σ, and no other bin by more than 0.99σ. No closure test with an independent generator has been made, and no control region constrains the proton mis-assignment of this sample (Section 8).

5.1 Closure under the nominal model

The pseudo-data are unfolded and compared with the A_C-smeared truth of the throw (Table 6). In FHC the χ^2 is well below the number of bins for every observable, and the integral ratios are 0.94–1.21. These ratios compare the unfolded result with the A_C-smeared truth, so the A_C suppression cancels and the departures from unity reflect the statistical realisation of the throw and any residual bias; the largest are in $\cos\theta_\mu$ (1.21) and $\delta\alpha_T$ (0.94). In RHC and combined the closure gives $\chi^2/\mathrm{ndf} \lesssim 1$ for every observable except $\cos\theta_\pi$ (7.3/4 in RHC, 5.2/4 combined). As in the inclusive analysis, the systematic terms are common to the pseudo-data and the reference, so the closure p-values cluster near unity and the statistical coverage is tested with ensembles.

5.2 Statistical coverage

Ten extractions were repeated on 100 statistically independent pseudo-data sets each, with the procedure of the inclusive ensembles (Sec. 17 of the technical supplement). Every member is a run-by-run Poisson throw of the central-value prediction carried through the whole chain: the universes are rebuilt from the throw and the result is unfolded. Each unfolded bin is compared with the fixed central-value truth smeared by that member's own A_C,

$$p_i \;=\; \frac{\hat{x}_i - (A_C\, x_{\mathrm{CV}})_i}{\sigma_i}.$$

Three of them span the validation-only products: W_{had} in two regions in FHC, the two-bin δp_T in FHC, and the two-bin p_n in the combined configuration, which also exercises the combination of the beam modes. The other seven are the differential results: θ_p and $\theta_{\pi p}$ in each configuration and $\theta_p \times \delta p_T$ in the combined configuration. For all 5600 input files the number of events read by the universe builder agrees with the entries on disk, so no member read an incomplete file.

Table 3: Proton-tagged pseudo-data ensembles, 100 members each. Pulls of the unfolded bins against the central-value truth smeared by each member's own A_C, in units of the statistical uncertainty (mean, width, fraction within 1σ and 1.96σ) and of the full framework uncertainty without the MCS term (width), the offset of the ensemble-mean integral from the reference integral (in brackets: in standard errors of the mean) and the member-to-member spread of the integral. Expected: mean 0, width 1, coverage 68% and 95% (`data_release/ensemble_1p_2026-10-09.tsv`, `ensemble_v1_2026-10-09.tsv`).

		statistical				full	integral	
Extraction	Config.	mean	width	68%	95%	width	offset	spread
W_{had} (2 regions)	FHC	−0.05	0.93	72%	96%	0.27	−0.4% (−0.5)	8.2%
δp_T (2 bins)	FHC	−0.06	1.09	64%	94%	0.40	−0.9% (−0.9)	9.3%
p_n (2 bins)	combined	−0.04	1.03	66%	95%	0.32	−0.1% (−0.2)	5.2%
θ_p	FHC	−0.08	1.01	70%	94%	0.41	−0.8% (−0.7)	11.3%
	RHC	−0.06	0.96	68%	96%	0.40	−1.0% (−1.0)	10.7%
	combined	−0.07	1.02	66%	96%	0.33	−0.7% (−1.2)	6.1%
$\theta_{\pi p}$	FHC	−0.03	1.00	70%	95%	0.42	0.0% (0.0)	9.3%
	RHC	−0.05	0.92	72%	97%	0.34	−0.4% (−0.5)	7.5%
	combined	−0.03	1.00	67%	95%	0.39	+0.3% (+0.2)	10.7%
$\theta_p \times \delta p_T$	combined	−0.04	1.04	66%	95%	0.42	−0.3% (−0.6)	5.0%

Against the statistical covariance the pull widths are 0.92–1.09, each known to about ±7%, and the 68% and 95% coverages are 64–73% and 94–97%. The widest case is the two-bin δp_T (width 1.09, 68% coverage 64%), within two standard deviations of nominal. The coverages pool the correlated bins of an extraction and are descriptive. Bin by bin, every mean is within 0.25 of zero and every width is 0.84–1.13. The narrowest, in the first RHC $\theta_{\pi p}$ bin, means that the statistical uncertainty of that bin is about 19% larger than the spread of its unfolded values, which is conservative. The statistical uncertainties of the proton-tagged results are therefore correctly sized.

Against the full covariance the widths are 0.27–0.42. The systematic part of the uncertainty is two to three times the statistical part, as in the inclusive analysis, and the ensembles do not test it, since every member is thrown from the same central-value model; that is the role of the alternative-model test below. The ensemble-mean integrals lie between 1.0% below and 0.3% above the reference, at most 1.2 standard errors of the mean, the small negative offset common to the inclusive ensembles (Sec. 13.2 of the inclusive analysis note). The other proton-tagged observables were not run individually.

5.3 Robustness to reweighted interaction-model variations

The pseudo-data are thrown from a reweighted interaction model and unfolded with the nominal response (Sec. 13.3 of the inclusive analysis note). The two variations of the inclusive analysis are used, GENIE multisim universe 545, a coherent variation of all interaction parameters, and the $\Delta \to N\pi$ decay-angle unisim, with the same universes and seeds; the weight applies to signal and background alike. W_{had}, the four TKI observables, $\cos\theta_\pi$ and p_p are extracted in FHC. The truth of the pseudo-data is that of the alternative model, so the closure measures the bias from the model mismatch, quoted in units of the per-bin uncertainty (Table 4).

Table 4: Alternative-model closure of the proton-tagged extractions, FHC: unfolded result against the alternative model's own truth. "ratio" is the mean unfolded/truth over bins holding at least 10% of the largest truth; "bias" is the largest per-bin deviation, over every bin, in units of the framework uncertainty of that bin (without the MCS term). $\theta_p \times \delta p_T$ is extracted in the combined configuration, the only one in which it is reported; its cells are numbered outer slice by outer slice (δp_T below and above 0.3 GeV/c), θ_p fastest.

Observable	Variation	ratio	max. deviation	bias (bin)
W_{had} (2 regions)	GENIE u545	0.74	52%	-0.69σ (1)
	$\Delta \to N\pi$ angular	1.27	67%	$+0.99\sigma$ (1)
δp_T (2 bins)	GENIE u545	0.68	32%	-1.08σ (1)
	$\Delta \to N\pi$ angular	1.01	1%	$+0.17\sigma$ (2)
$\delta\alpha_T$ (2 bins)	GENIE u545	0.83	28%	-0.67σ (1)
	$\Delta \to N\pi$ angular	1.02	10%	$+0.24\sigma$ (2)
$\delta\phi_T$ (3 bins)	GENIE u545	0.80	33%	-0.57σ (2)
	$\Delta \to N\pi$ angular	1.23	50%	$+0.89\sigma$ (2)
p_n (2 bins)	GENIE u545	0.95	28%	-1.04σ (1)
	$\Delta \to N\pi$ angular	1.09	21%	$+0.25\sigma$ (2)
$\cos\theta_\pi$	GENIE u545	0.98	12%	-0.41σ (1)
	$\Delta \to N\pi$ angular	0.96	25%	-0.59σ (3)
p_p (3 bins)	GENIE u545	0.90	13%	-0.28σ (1)
	$\Delta \to N\pi$ angular	1.07	16%	$+0.38\sigma$ (1)
θ_p	GENIE u545	0.82	28%	-0.79σ (2)
	$\Delta \to N\pi$ angular	1.12	25%	$+0.68\sigma$ (3)
$\theta_{\pi p}$	GENIE u545	0.89	15%	-0.28σ (1)
	$\Delta \to N\pi$ angular	1.04	35%	$+0.65\sigma$ (1)
$\theta_p \times \delta p_T$	GENIE u545	0.95	30%	-0.88σ (2)
	$\Delta \to N\pi$ angular	0.98	47%	-0.58σ (3)

Under the $\Delta \to N\pi$ variation no bin exceeds 0.99σ, although the two W_{had} regions and the three $\delta\phi_T$ bins return 23–27% high on average. Under the coherent GENIE variation the low δp_T and p_n bins, the small-imbalance region closest to quasi-elastic kinematics, return 32% and 28% below the alternative truth, -1.08 and -1.04σ, while that truth moved by only 2%. The shift therefore enters through the response and the background, through the change the varied model makes to the composition of the proton-tagged sample and to the proton-tagging efficiency, which the nominal response does not follow. The inclusive analysis shows a bias of this size in one sparse bin only (Sec. 13.3 of the inclusive analysis note). The uncertainty covers the shift at about one standard deviation, so the extraction does not fail, but the small-imbalance bins of the TKI observables are model-sensitive at the level of their own uncertainty, and the covariance contains no term for this model dependence.

The results stay within 0.9σ in every bin. Under the GENIE variation all four θ_p bins return low, 0.72–0.94 of the alternative truth, while that truth changed by less than 1%; the largest bias is -0.79σ, in the second bin. The integral of the FHC θ_p extraction varies by 11% from one pseudo-data set to the next (§5.2), so a single throw cannot separate a model effect of this size from a fluctuation; in either case every bin stays below 1σ. Under the angular variation the first $\theta_{\pi p}$ bin returns 35% high ($+0.65\sigma$). The two low-imbalance cells of $\theta_p \times \delta p_T$ return 0.83 and 0.70 of the alternative truth under the GENIE variation (-0.66 and -0.88σ), in the direction of the low δp_T bin of the one-dimensional result but within 1σ. Under the angular variation the cell with $\delta p_T > 0.3$ GeV/c and $\theta_p < 0.8$ rad returns 0.53 of the alternative truth, the 47% deviation of the table, which is -0.58σ against the large uncertainty of that cell.

6 Results

6.1 Total cross section

The total cross section is extracted as a single bin, as in the inclusive analysis (Sec. 14.6 of the inclusive analysis note): nothing is regularised and $A_C = 1$, so the one-bin result is the physical total in the measured phase space and does not depend on an observable (Table 5). A closed-form evaluation on the central-value prediction reproduces the one-bin extraction of that prediction to four digits (0.6269, 0.5506 and 0.5831 for FHC, RHC and combined). The extracted RHC/FHC ratio is 0.873, against 0.878 for the tune.

Table 5: Proton-tagged total cross sections (10^{-38} cm^2/Ar) in the measured phase space, averaged over the $\nu_\mu + \bar{\nu}_\mu$ flux above 60 MeV, which includes the flux below the pion-production threshold, from the one-bin extraction on pseudo-data. N_{sel} and N_{bkg} are the predicted selected count and background (ν-background + EXT + dirt) of the central-value prediction and ε the selection efficiency; "stat" is the data-statistical term, "syst" all other terms and "total" their quadrature sum; "truth" is the realised truth of the throw and $\chi^2/1$ the closure against it.

Config	N_{sel}	N_{bkg}	ε [%]	σ	stat [%]	syst [%]	total [%]	truth	σ/truth	$\chi^2/1$
FHC	677	328	12.1	0.618 ± 0.251	7.5	39.9	40.6	0.639	0.968	0.007
RHC	834	420	12.1	0.522 ± 0.224	7.3	42.2	42.9	0.532	0.981	0.002
Combined	1 510	749	12.1	0.563 ± 0.228	5.2	40.2	40.5	0.578	0.975	0.004

The flux-averaged FHC generator totals in the proton-tagged phase space are 0.50 (GENIE), 0.65 (NuWro), 0.74 (GiBUU) and 0.78×10^{-38} cm^2/Ar (NEUT), a factor 1.6 between the lowest and the highest, around the tune value of 0.63.

6.2 Proton angles θ_p and $\theta_{\pi p}$

θ_p uses the stored proton direction; $\theta_{\pi p}$ is written by a reprocessing of the proton-tagged tree that leaves every other branch unchanged. Both are binned in four bins and unfold well, with closure χ^2 of 5.4/4, 0.06/4 and 2.1/4 for θ_p and 0.30/4, 0.90/4 and 0.32/4 for $\theta_{\pi p}$ in FHC, RHC and combined (Fig. 1). $\theta_{\pi p}$ retains the most shape: the trace of A_C is 2.2–2.3, with three components measured at a signal-to-noise ratio above one in FHC and RHC, at the price of a last bin whose total uncertainty reaches 82% (FHC) and 112% (RHC). θ_p has a trace of 2.3 in FHC and 2.0 in RHC (2.7 combined) and a largest per-bin uncertainty of 41% (FHC) and 46% (RHC); its smallest diagonal passes the 0.68 criterion by 0.005 in RHC. Both distributions are strongly forward peaked, and the generators differ mainly in normalisation, so these observables discriminate between models through their integrals. Ensembles of 100 pseudo-data sets in each configuration give statistical pull widths of 0.92–1.02, and under the two alternative models no bin moves by more than 0.79σ (§5.2 and §5.3). Their binnings follow the rule of Sec. 11.3 of the inclusive analysis note; the screening of the two-dimensional pairs is described in Sec. 10 of the technical supplement.

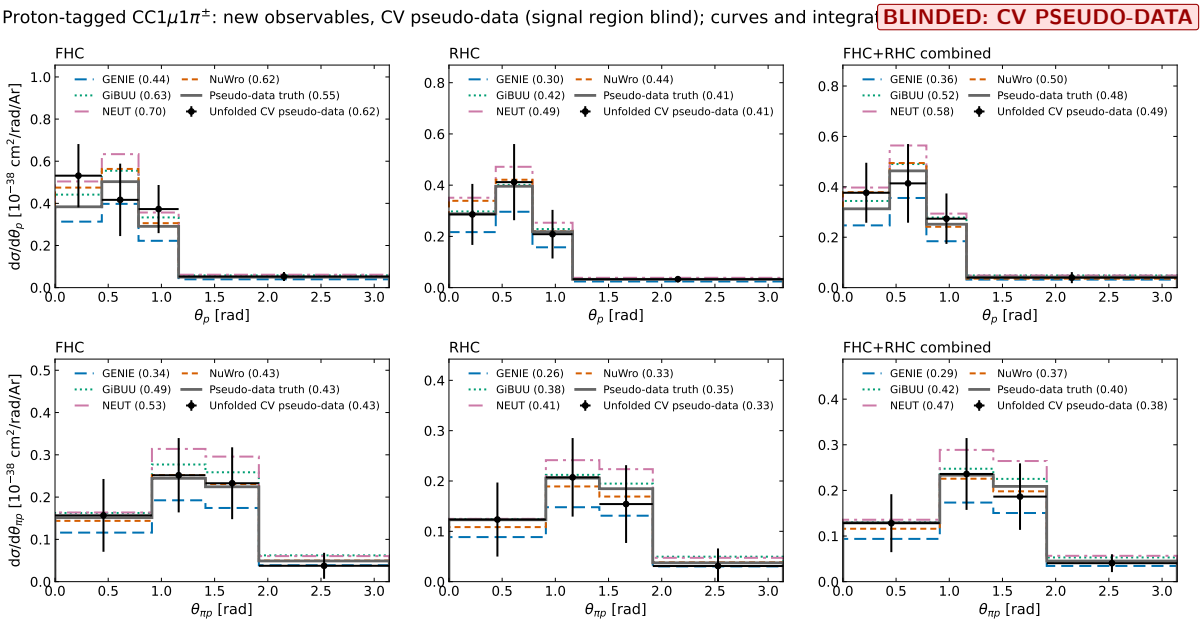


Figure 1: The two proton angles in the three configurations: θ_p (top row) and $\theta_{\pi p}$ (bottom row), for FHC, RHC and combined. Points are the unfolded pseudo-data with the total uncertainty, the grey histogram is the A_C-smeared pseudo-data truth, and the four coloured histograms are the generator predictions; the integrated cross sections are given in the legends.

6.3 Two-dimensional $\theta_p \times \delta p_T$

The double-differential cross section $d^2\sigma/d\theta_p\, d\delta p_T$ is reported in the combined configuration only. It is binned in two regions of θ_p (edges 0, 0.8, π rad) inside two of δp_T (0, 0.3 GeV/c and above); the four analysis bins are ordered outer slice by outer slice and extracted with the same machinery as the one-dimensional results. It is one of eight pairs screened (Sec. 10 of the technical supplement). Combining the beam modes raises the conditioning $s_{\min}/s_1$ of its A_C from 0.584 (FHC) and 0.491 (RHC) to 0.802, through the larger sample and the shared flux; the FHC and RHC extractions are Not reported. The combined result has a bin-averaged data-statistical term of 12.9% and per-bin total uncertainties of 27–135%; it closes against the A_C-smeared truth with $\chi^2 = 1.71/4$ and per-bin ratios of 0.62–1.32 (Figs. 2 and 3). Because it uses δp_T, it depends on the muon momentum and carries the muon-momentum-scale term, which moves its cells by at most 24% of their value and 0.24σ and leaves its integral unchanged; its low-imbalance bins share the model sensitivity of the one-dimensional δp_T (§5.3). Over 100 pseudo-data sets its statistical pull width is 1.04, and under the two alternative models no cell moves by more than 0.88σ (§5.2 and §5.3). Its covariance, A_C and curves are released as `1p_COMB_thetap_dpt2d`.

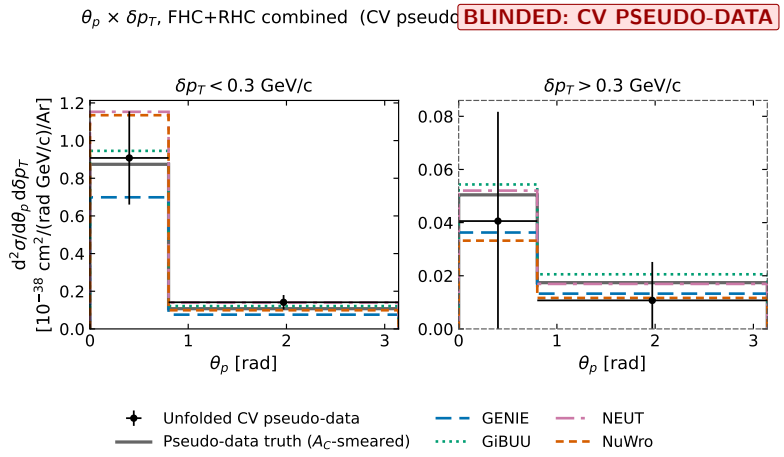


Figure 2: $d^2\sigma/d\theta_p\, d\delta p_T$ in the combined configuration, as slices in δp_T. Points are the unfolded pseudo-data with the total uncertainty, the grey histogram is the A_C-smeared truth of the pseudo-data and the coloured histograms are the generator predictions, smeared with A_C. Each slice has its own vertical scale.

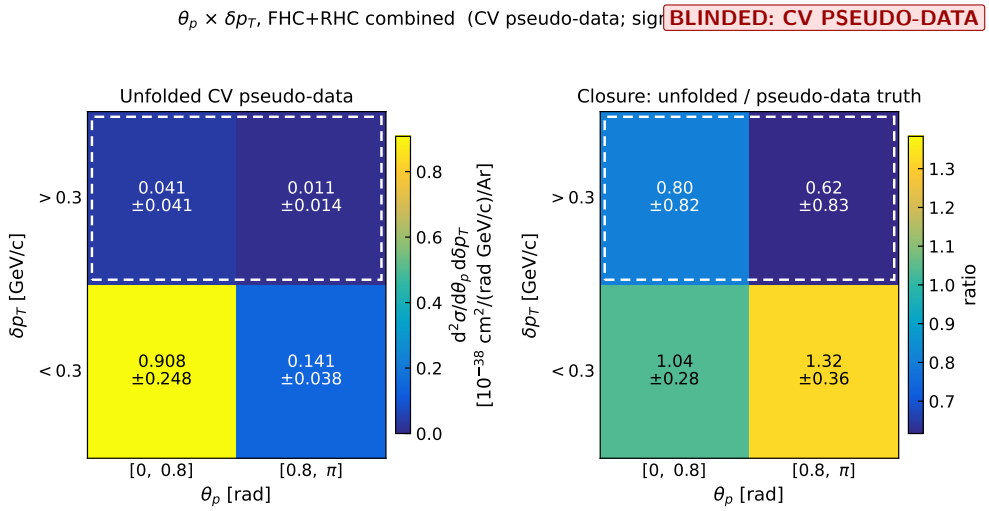


Figure 3: The same result as a map: $d^2\sigma/d\theta_p\, d\delta p_T$ per cell with its total uncertainty (left) and the ratio to the A_C-smeared truth (right). Cells are drawn with equal size and labelled by their ranges.

6.4 Differential cross sections

The differential results of this section carry their full covariances and are released, and the limitations of Section 8 apply to them. Table 6 gives the integrated cross section of each unfolded result and its closure against the A_C-smeared truth of the pseudo-data, and Figs. 4–6 show the six observables specific to this sample.

Table 6: Proton-tagged results from pseudo-data. σ_{int} is the integrated unfolded cross section over the measured phase space (10^{-38} cm^2/Ar), "ratio" its ratio to the integral of the A_C-smeared truth, and χ^2/n the bin-by-bin closure against that truth (n bins). The upper block lists the observables specific to the sample, the lower block the muon and pion observables of the inclusive analysis; the proton angles are in §6.2.

	FHC			RHC			Combined		
Observable	σ_{int}	ratio	χ^2/n	σ_{int}	ratio	χ^2/n	σ_{int}	ratio	χ^2/n
W_{had}	0.536	0.98	0.05/2	0.411	0.95	0.27/2	0.466	0.96	0.06/2
δp_T	0.499	1.04	0.33/2	0.496	1.17	0.89/2	0.649	1.18	0.50/2
$\delta\alpha_T$	0.473	0.94	0.06/2	0.430	1.04	0.14/2	0.447	0.99	0.01/2
$\delta\phi_T$	0.508	1.07	0.40/3	0.497	1.03	0.28/3	0.568	1.04	0.23/3
p_n	0.486	0.99	1.15/2	0.405	0.99	1.46/2	0.466	1.05	2.15/2
p_p	0.484	1.00	0.02/3	0.397	0.99	0.20/3	0.445	1.01	0.12/3
p_μ	0.501	1.04	0.52/6	0.355	0.92	0.48/6	0.413	0.97	0.90/6
p_π	0.496	0.97	0.01/2	0.437	1.06	1.23/2	0.461	1.01	0.34/2
$\cos\theta_\mu$	0.537	1.21	2.77/8	0.308	0.91	1.99/8	0.367	0.95	4.44/8
$\cos\theta_\pi$	0.402	1.00	1.46/4	0.373	1.08	7.28/4	0.397	1.04	5.25/4
$\theta_{\mu\pi}$	0.464	1.00	0.39/5	0.382	0.99	0.04/5	0.423	1.02	0.20/5

Each A_C changes the normalisation by its own factor, so the integrated cross sections depend on the observable (0.37–0.65 in the combined configuration, against a one-bin total of 0.563) and are not independent measurements of one quantity; the RHC/FHC ratio is taken from the one-bin totals. The combined result lies between the FHC and RHC results for nine of the eleven extractions and slightly above both for δp_T and $\delta\phi_T$, which is allowed for an independent extraction with its own A_C and is well inside the uncertainties.

The generator predictions are computed at truth level from the event-level final state of each generator (GENIE [2, 4] and GiBUU [6] natively, NuWro [5] and NEUT [10, 7] from their event vectors), with the proton-tagged signal definition and the observable definitions of the selection (Sec. 9.1 of the technical supplement), and combined over the ν_μ and $\bar{\nu}_\mu$ flux of the configuration; each one-dimensional prediction integrates to the proton-tagged total of its generator. All curves, the tune and the pseudo-data truth included, are smeared with A_C (Sec. 9.2 of the technical supplement). In the reported binnings the generators differ mostly in normalisation. The tune and all four generators are compatible with the unfolded pseudo-data in every panel and configuration: with the official covariance the largest χ^2 is 4.4 on two bins (GENIE, combined p_n), GENIE having the lowest normalisation, and all others are below 3.6. The present exposure therefore does not discriminate between the generators' treatments of Fermi motion and final-state interactions. A comparison in which data and prediction used different axes would show shape differences of a factor of several, so the frame is stated with every transverse quantity.

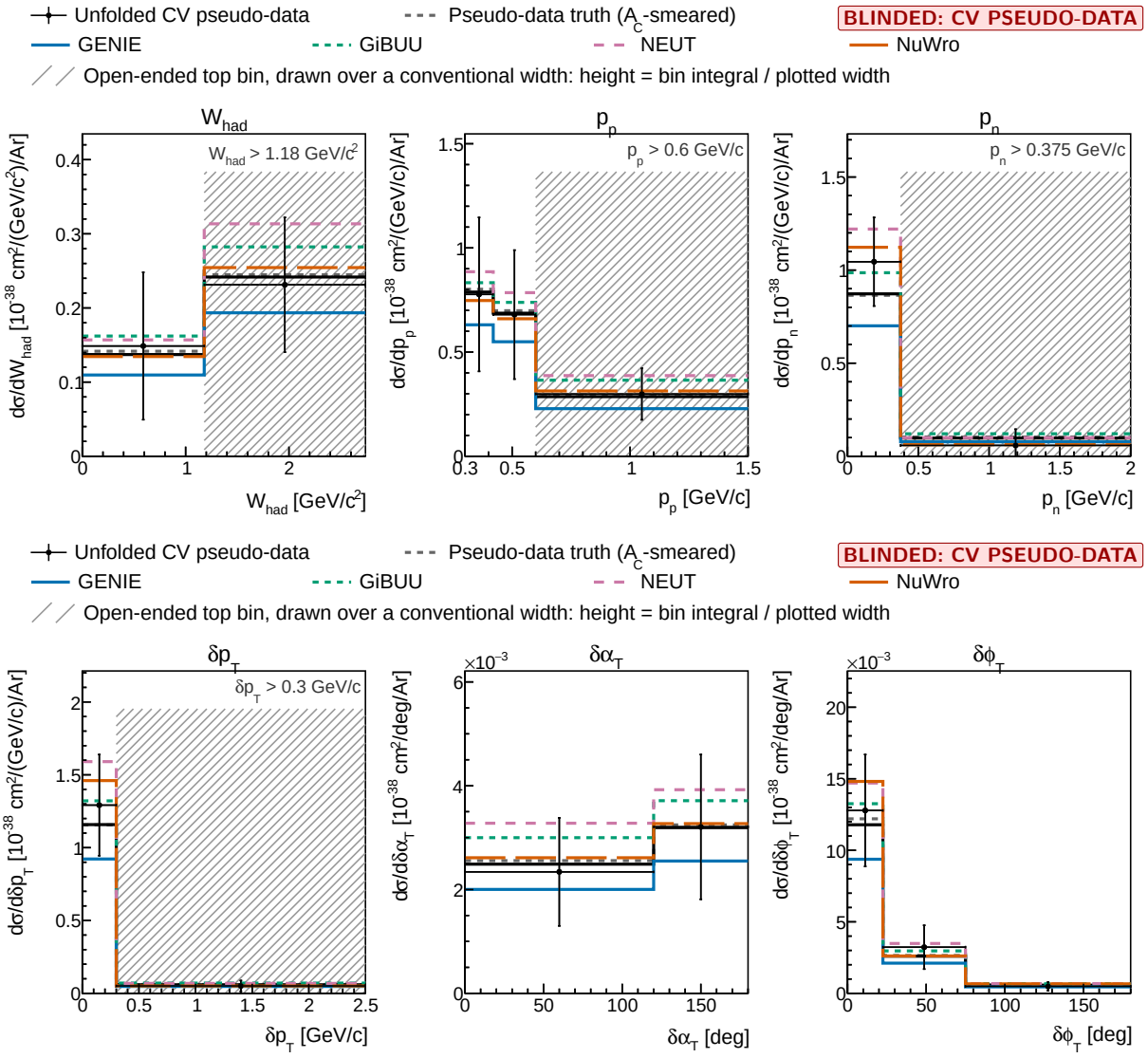


Figure 4: Proton-tagged differential cross sections, FHC: W_{had}, p_p and p_n (top) and δp_T, $\delta\alpha_T$ and $\delta\phi_T$ (bottom). Unfolded pseudo-data (points) are compared with the A_C-smeared pseudo-data truth (grey dashed), the MicroBooNE tune [3] (black) and the GENIE (blue), GiBUU (green), NEUT (purple) and NuWro (orange) predictions for the same phase space. The last bin of each open-ended observable is hatched and labelled with its range; its height is the bin integral divided by the plotted width. $\delta\alpha_T$ rises towards 180°, the signature of final-state interactions.

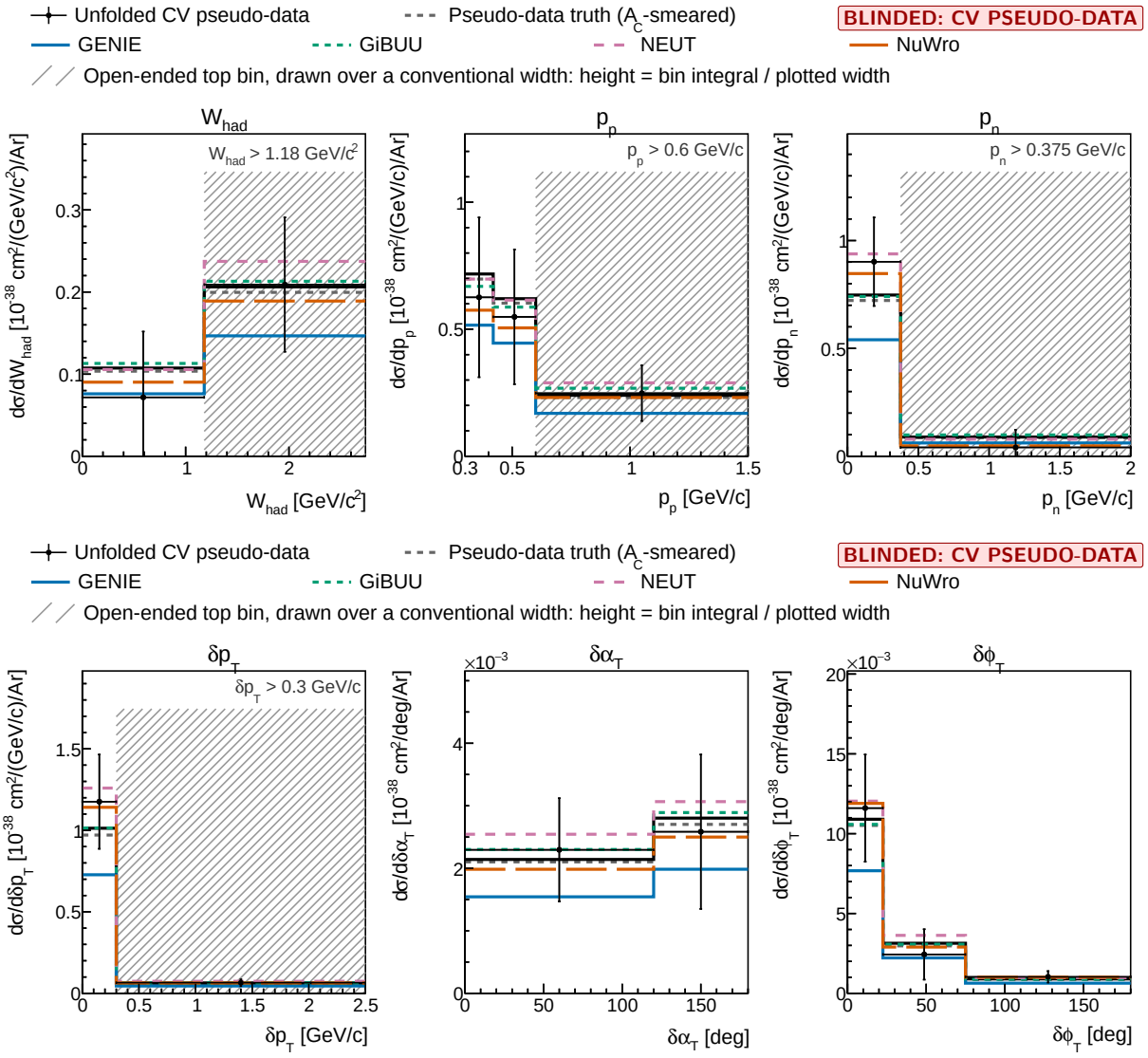


Figure 5: Proton-tagged differential cross sections, RHC: W_{had}, p_p and p_n (top) and δp_T, $\delta\alpha_T$ and $\delta\phi_T$ (bottom). Unfolded pseudo-data (points) are compared with the A_C-smeared pseudo-data truth (grey dashed), the MicroBooNE tune (black) and the GENIE (blue), GiBUU (green), NEUT (purple) and NuWro (orange) predictions for the same phase space. The last bin of each open-ended observable is hatched and labelled with its range; its height is the bin integral divided by the plotted width.

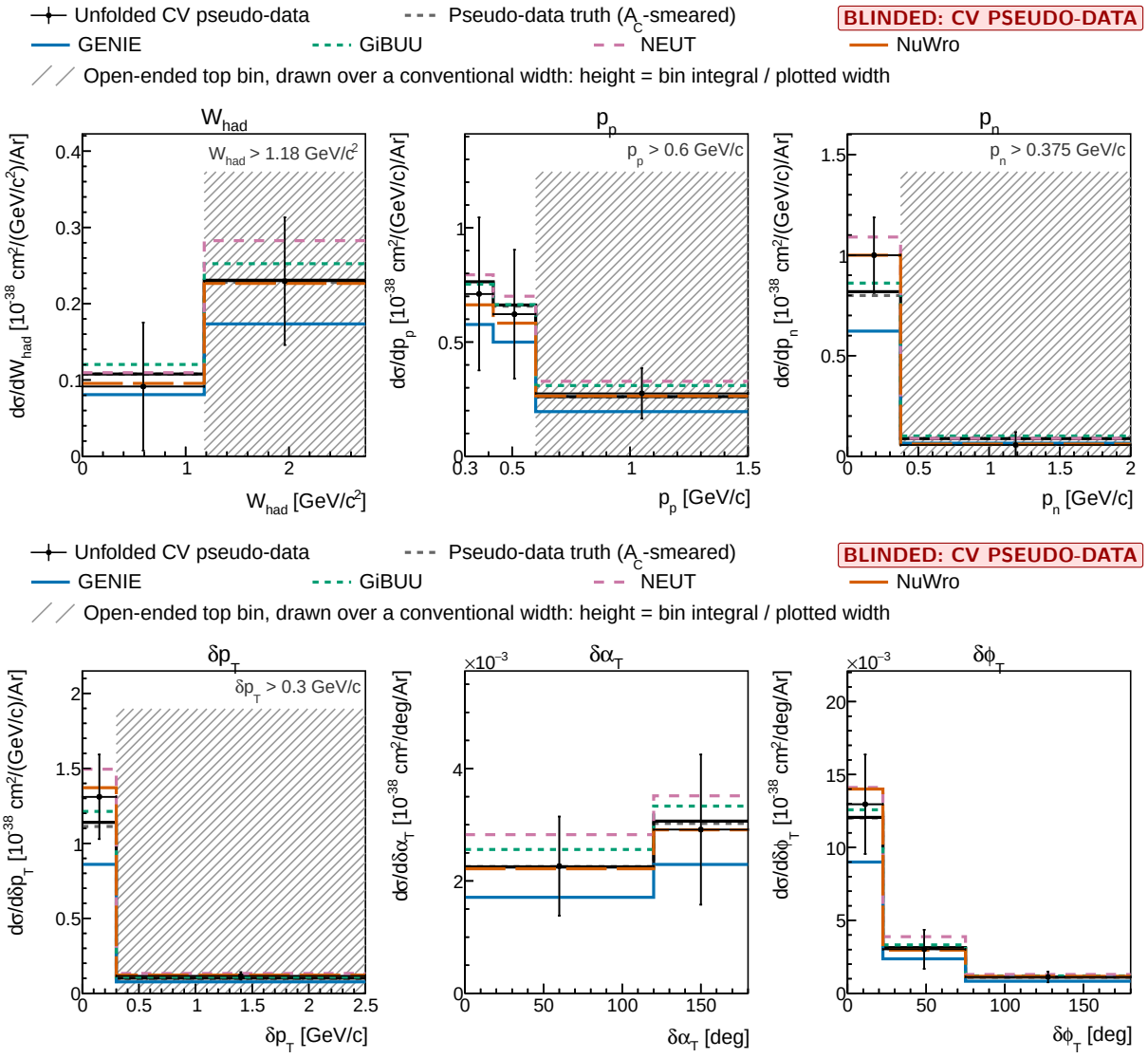


Figure 6: Proton-tagged differential cross sections, the combined configuration: W_{had}, p_p and p_n (top) and δp_T, $\delta\alpha_T$ and $\delta\phi_T$ (bottom). Unfolded pseudo-data (points) are compared with the A_C-smeared pseudo-data truth (grey dashed), the MicroBooNE tune (black) and the GENIE (blue), GiBUU (green), NEUT (purple) and NuWro (orange) predictions for the same phase space. The last bin of each open-ended observable is hatched and labelled with its range; its height is the bin integral divided by the plotted width.

7 Products not reported

No differential $W_{\pi p}$ result is reported. The topology-level one-bin total cross section of the proton-tagged sample remains measurable (§6.1), but it is not described as a $W_{\pi p}$ measurement. Of the two-dimensional pairs screened, those built on $W_{\pi p}$ and $\theta_{\pi p} \times \delta p_T$ cannot be inverted or carry no usable information, and the FHC and RHC extractions of $\theta_p \times \delta p_T$ are not shape measurements (Sec. 10 of the technical supplement); none of them is reported.

7.1 The differential $W_{\pi p}$ shape is not reported

The pion–proton invariant mass would show the $\Delta(1232)$ directly and motivated the proton-tagged subsample. At the present exposure its shape cannot be measured with any binning tried. The six-bin scheme that resolves the resonance region has a highest bin with a 2% diagonal,

so nearly every event reconstructed there comes from another true bin. Two bins restore the response but leave no shape: the best split, at 1.20 GeV/c^2, passes the 0.50 criterion by 0.003, and its second Wiener filter factor is 0.29 in FHC and 0.27 in RHC. A closure test against the A_C-smeared truth cannot change this, since it tests whether the response was inverted correctly and cannot restore information the response does not contain. The six-bin extractions remain in the result directory, flagged in `index_extractions.tsv`, and their closure and generator curves are shown as diagnostics in Sec. 8.1 of the technical supplement; they are not results. The sample supports the topology-level one-bin total and, as a validation-only product, W_{had} in two regions.

8 Limitations

No control region constrains the dominant proton-candidate misidentification of this sample. The multi-π region of the inclusive analysis constrains the proton→pion direction, while the 12.6% of wrong candidates come mainly from the pion→proton direction (§2.1); an inverted-proton-PID region, which would separate the two, has not been defined. The distributions whose migrations are dominated by the proton-candidate assignment, θ_p, $\theta_{\pi p}$ and $\theta_p \times \delta p_T$, carry no uncertainty for the rate of wrong candidates.

Two further limitations concern the transverse observables. The first is the validity of the reconstruction: W_{had} and p_n use a residual-mass prescription derived for quasi-elastic kinematics, and it has not been shown that in pion production the reconstructed quantities follow the hadronic invariant mass and the initial-nucleon momentum they are named for. The second is model sensitivity: the low δp_T and p_n bins move by 1.04–1.08σ under the coherent GENIE variation while their truth moves by 2% (§5.3), and the covariance contains no term for this model dependence.

9 Summary

The proton-tagged selection adds to the inclusive selection a contained proton candidate above 0.3 GeV/c, at 12.1% efficiency and 50–52% purity; the candidate is the leading true proton in 82% of selected signal events. The one-bin total cross section, $d\sigma/d\theta_p$ and $d\sigma/d\theta_{\pi p}$ in the three configurations, and $d^2\sigma/d\theta_p\, d\delta p_T$ in the combined configuration and no differential $W_{\pi p}$ result is reported. The extraction closes, and ensembles give correctly sized statistical uncertainties. The low-imbalance bins of δp_T and p_n are model-sensitive at the level of their uncertainty, and no control region constrains the proton mis-assignment. The signal region is blind for every proton-tagged result.

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